

Comparative Analysis of Unsupervised Clustering Algorithms on Housing Data

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Abstract—No one can deny that unsupervised learning has an important role in data exploration and pattern discovery, specifically when labeled data is unavailable. This paper presents a comparative analysis of many clustering algorithms applied to a housing dataset containing socioeconomic and demographic features. The main goal of this work is to compare and evaluate the performance of different unsupervised learning techniques on the same dataset.

Four clustering algorithms were implemented and compared: K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models (GMM). Before model training, the dataset was preprocessed through data cleaning, standardization, and outlier treatment using the Interquartile Range (IQR) method. Principal Component Analysis (PCA) was used for two-dimensional visualization of the generated clusters.

The experimental results showed that K-Means and GMM produced more structured and balanced clusters, while Agglomerative Clustering provided hierarchical group separation. DBSCAN demonstrated strong capabilities in detecting noise and outliers but showed sensitivity to parameter selection and generated fewer clearly separated clusters on this dataset. Visual analysis using PCA highlighted differences in cluster compactness and separation among the models.

This comparative study demonstrates that clustering performance strongly depends on data distribution and algorithm characteristics. The results provide valuable insights into the strengths and limitations of different clustering approaches for exploratory data analysis.

Keywords—Unsupervised Learning, Clustering, K-Means, DBSCAN, Agglomerative Clustering, Gaussian Mixture Model, PCA, Housing Data Analysis, Machine Learning

I. INTRODUCTION

In our current era, the increase in data has made analysis and knowledge discovery techniques extremely important. In many real-world applications, labeled data is unavailable or expensive to obtain, making unsupervised learning essential for exploring hidden structures and patterns within datasets. Clustering is one of the most widely used unsupervised learning techniques, aiming to group similar observations into meaningful clusters based on their characteristics [1].

Clustering algorithms are widely used in many fields, such as healthcare, finance, marketing, cybersecurity, and geographic analysis. These methods help identify hidden relationships between data and support decision-making processes through exploratory data analysis. However, the performance of clustering algorithms depends heavily on the nature of the

dataset, the preprocessing techniques employed, and the choice of parameters [2].

This study focused on the comparative analysis of multiple clustering algorithms applied to a housing dataset containing socioeconomic and demographic characteristics. The research aimed to evaluate the effectiveness of different unsupervised learning techniques in identifying significant clusters within the data. Four clustering algorithms were applied and compared: K-Means, DBSCAN, Agglomerative Clustering, and Gaussian mixture models (GMMs).

Before using these clustering models, several preprocessing steps were performed, including data cleaning and standardization, and outlier treatment using the Interquartile Range (IQR) method. Additionally, principal component analysis (PCA) [3] was used to visualize the clusters in two-dimensional space and facilitate the interpretation of the results [4].

The main added value of this work lies in the comparative evaluation of clustering methods on the same dataset using internal evaluation metrics and visual analysis. The study also highlights the strengths and weaknesses of each algorithm in terms of clustering quality, noise resistance, and criteria sensitivity.

The remainder of this paper is organized as follows: Section II presents the related work and theoretical background of the clustering algorithms. Section III describes the methodology and preprocessing steps. Section IV presents the implementation details and experimental setup. Section V discusses the results and comparative analysis of the models. Finally, Section VI concludes the paper and presents future research perspectives.

II. STATE OF THE ART

A. Related Work

A recent study titled "Unsupervised Learning: A Comparative Analysis of Clustering Techniques on High-Dimensional Data" provided a comparative evaluation of K-Means, DBSCAN, and spectral clustering algorithms on high-dimensional datasets. This study analyzed the impact of dimensionality reduction techniques, such as PCA, t-SNE, and UMAP, on clustering quality. The results showed that preprocessing of the UMAP algorithm significantly improved clustering performance, while spectral clustering achieved better results

on complex data structures. The researchers also concluded that K-Means exhibits high computational efficiency, while DBSCAN excels in detecting irregular clusters and noisy data [5].

Another comparative study, titled "A Comparative Analysis of Clustering Algorithms," analyzed several clustering techniques, including K-Means, Gaussian mixture models, DBSCAN, and Agglomerative Clustering, on datasets with varying density distributions. The study compared the different algorithms based on clustering accuracy, dataset complexity, scalability, and sensitivity to data distribution. The final results showed that DBSCAN is effective on datasets with unclear density structures and noisy observations, while Agglomerative Clustering provides better identification of hierarchical data structures. However, the researchers concluded that no single clustering algorithm is the best and most stable for all datasets and recommended combining preprocessing techniques with multiple clustering approaches to improve clustering quality [6].

Another study, titled "Exploring and Comparing Unsupervised Clustering Algorithms," presented a comparative analysis of the K-Means algorithm and Gaussian Mixture Models (GMMs) for unsupervised clustering applications. The paper demonstrated the impact of R-developed tools on visualizing and comparing clustering results using simulated datasets and the Iris dataset. The research concluded that the K-Means algorithm achieves efficient clustering through center-based partitioning, while GMMs offer more flexible probabilistic clustering capable of handling uncertainty and overlapping data distributions. This means that comparative analysis and visualization techniques are crucial for understanding clustering behavior and improving exploratory data analysis [7].

These previous works demonstrate the importance of comparative studies in unsupervised learning. Motivated by these findings, this study evaluates and compares K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models on a housing dataset in order to analyze their clustering behavior, robustness, and visualization performance.

B. Presentation of the Algorithms

Clustering is a key technique in unsupervised machine learning. It aims to group similar observations into clusters without using labeled data. Clustering techniques are widely used in many fields, such as healthcare, finance, customer segmentation, environmental analysis, and housing data exploration. The goal of clustering is to discover hidden structures and meaningful patterns within complex datasets.

Among the most commonly used clustering algorithms, K-Means remains one of the most popular methods due to its simplicity and computational efficiency. K-Means partitions the dataset into a predefined number of clusters by minimizing the distance between observations and cluster centroids. Despite its effectiveness on well-separated datasets, the algorithm is sensitive to outliers and requires the number of clusters to be specified in advance [8]. The objective function of K-Means

aims to minimize the distance between observations and their corresponding cluster centroids:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2 \quad (1)$$

where C_i represents cluster i , μ_i is the centroid of the cluster, and x represents observations belonging to the cluster.

One of the algorithms we used is DBSCAN, a density-based clustering algorithm that groups data points that are closely packed together and marks outliers as noise based on their density in the feature space. It identifies clusters as dense regions in the data space separated by areas of lower density. Unlike K-Means or hierarchical clustering, which assume clusters are compact and spherical, DBSCAN can detect clusters of arbitrary shapes and is more robust to noise and outliers [9]. DBSCAN defines clusters based on density connectivity using the neighborhood radius parameter ε :

$$N_\varepsilon(p) = \{q \in D \mid \text{dist}(p, q) \leq \varepsilon\} \quad (2)$$

where $N_\varepsilon(p)$ represents the neighborhood of point p , and $\text{dist}(p, q)$ is the distance between observations p and q .

We also used Agglomerative Clustering, which is a hierarchical clustering approach that progressively merges observations into larger clusters based on similarity measures. This method provides a hierarchical representation of the data structure through dendrograms and can reveal nested cluster relationships. Nevertheless, hierarchical clustering can become computationally expensive for large datasets [10]. Agglomerative Clustering progressively merges clusters based on similarity measures. The distance between clusters can be calculated using linkage methods such as single linkage:

$$D(A, B) = \min_{a \in A, b \in B} d(a, b) \quad (3)$$

where A and B are two clusters and $d(a, b)$ represents the distance between observations.

Finally, we used a Gaussian Mixture Model (GMM), which is a probabilistic model that represents data as a combination of several Gaussian distributions, each with its own mean and variance, weighted by a mixing coefficient. GMMs are commonly used for clustering and density estimation since they can capture complex, multimodal distributions where data points may naturally group around different centers rather than a single mean [11].

Gaussian Mixture Models represent the data distribution as a weighted combination of multiple Gaussian distributions:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k) \quad (4)$$

where π_k represents the mixing coefficient, μ_k is the mean vector, and Σ_k is the covariance matrix of the Gaussian component.

III. METHODOLOGY

A. Dataset

The dataset we used in this study is a housing dataset containing numerical data related to socioeconomic and demographic characteristics. Our main objective with this dataset is to represent different residential areas using variables related to income, housing conditions, and population statistics.

The original dataset contained the following variables: *MedInc*, *HouseAge*, *AveRooms*, *AveBedrms*, *Population*, *AveOccup*, *Latitude*, *Longitude*, and *MedHouseVal*. In this study, the variables *Latitude*, *Longitude*, and *MedHouseVal* were removed manually from the original file in order to focus the clustering process on the remaining socioeconomic and housing-related attributes.

The final dataset used for clustering contained the following features:

- *MedInc*
- *HouseAge*
- *AveRooms*
- *AveBedrms*
- *Population*
- *AveOccup*

The dataset was perfectly suited to distance-based clustering algorithms because all the variables were numerical.

B. Data Preprocessing

We preprocessed the data to improve data quality and prepare the dataset for cluster analysis. We applied several preprocessing steps before training the models.

The first thing we do is check for missing values to ensure data consistency. We found no missing values during the analysis.

Next, we applied feature scaling using the *StandardScaler* method from the Scikit-learn library [12]. Standardization transformed the variables to have a mean close to zero and a standard deviation close to one. This step was essential because clustering algorithms such as K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models rely heavily on distance calculations.

After that, we detected and processed outliers using the Interquartile Range (IQR) method [13]. This process was important for minimizing their impact on clustering performance, especially for distance-based algorithms. The preprocessing stage significantly improved the stability and quality of the clustering results.

In addition, we used principal Component Analysis (PCA) for dimensionality reduction and visualization purposes. PCA transformed the dataset into two principal components to facilitate the graphical representation of clusters generated by the different algorithms.

C. Model Selection

Four clustering algorithms were selected for comparative analysis in this study:

- K-Means

- DBSCAN
- Agglomerative Clustering
- Gaussian Mixture Models (GMM)

We chose the K-Means algorithm for its simplicity, computational efficiency, and widespread use in unsupervised learning tasks. We also chose the DBSCAN algorithm for its ability to detect noisy points and identify irregularly shaped clusters. Agglomerative Clustering was included to analyze hierarchical relationships between observations, while we selected Gaussian Mixture Models because of their probabilistic clustering capabilities and flexibility in handling overlapping clusters.

We evaluated the model using internal clustering evaluation metrics, including the Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. In addition, we used also PCA visualizations to compare the spatial distribution and separation of clusters generated by each algorithm.

IV. APPLICATION AND IMPLEMENTATION

A. Implementation

For the implementation of the clustering models we used Python programming language within a Jupyter Notebook environment. The workflow included data loading, exploratory data analysis, preprocessing, clustering model training, evaluation and visualization.

we began the implementation process with importing the housing dataset and performing exploratory data analysis to understand the structure and distribution of the variables. we generated histograms, boxplots, scatter plots, and correlation matrices to identify data characteristics and potential outliers [14].

After the exploratory phase, we applied preprocessing techniques, for feature scaling we used *StandardScaler* and for outlier treatment we used the Interquartile Range (IQR) method. These preprocessing steps improved the stability of distance-based clustering algorithms.

Let's now move on to the clustering algorithms that were implemented sequentially. the first one is K-Means clustering which applied using the Elbow Method to determine the optimal number of clusters. the second one is DBSCAN which was implemented using different values of the *eps* parameter to analyze cluster density and noise detection behavior. the third one is Agglomerative Clustering which also was applied to study hierarchical grouping relationships, finally we used Gaussian Mixture Models (GMM) for probabilistic clustering analysis.

To evaluate clustering quality, we calculated several internal evaluation metrics, including the Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index. we also used Principal Component Analysis (PCA) to reduce the dimensionality of the dataset and visualize the generated clusters in two-dimensional space [15].

B. Tools and Libraries

Several Python libraries and tools were used during the implementation phase:

- **Python:** Main programming language used for implementation [16].
- **Pandas:** Used for data loading, cleaning, and manipulation [17].
- **NumPy:** Used for numerical operations and array processing [18].
- **Matplotlib:** Used for data visualization and plotting graphs [20].
- **Seaborn:** Used for advanced statistical visualizations such as heatmaps and boxplots [19].
- **Scikit-learn (sklearn):** Main machine learning library used for preprocessing, clustering algorithms, PCA, and evaluation metrics [21].
- **Jupyter Notebook:** Development environment used to execute and organize the experiments interactively [22].

The Scikit-learn library provided efficient implementations of the clustering algorithms used in this study, including K-Means, DBSCAN, Agglomerative Clustering, Gaussian Mixture Models, PCA, and clustering evaluation metrics.

In addition, we published on **GitHub** [23] the complete source code of the project, including data preprocessing, clustering implementations, evaluation metrics, and visualizations, to ensure reproducibility and facilitate future improvements and experimentation.

V. RESULTS AND DISCUSSION

A. Evaluation Metrics

For evaluate clustering quality, we used many internal evaluation metrics in this study. these metrics helped measure cluster compactness, separation quality, and overall clustering performance without requiring labeled data.

The Silhouette Score is a metric used to evaluate the quality of clustering results. It measures how similar each data point is to its own cluster compared to other clusters, helping assess how well the data has been grouped. This score is widely used to evaluate clustering algorithms like K-Means. In this study we used it to measures how similar observations are to their own cluster compared to other clusters. A higher Silhouette Score indicates better cluster separation and more compact clustering structures [24].

The Davies-Bouldin Index is a well-known metric to perform the same - it enables one to assess the clustering quality by comparing the average similarity between the pairwise most similar clusters. we used it to evaluates the average similarity between clusters based on intra-cluster distances and inter-cluster distances. Lower values of the Davies-Bouldin Index indicate better clustering performance and stronger separation between clusters [25].

The Calinski-Harabasz index (CH) is one of the clustering algorithms evaluation measures. It is most commonly used to evaluate the goodness of split by a K-Means clustering algorithm for a given number of clusters. we used it as well to measures the ratio between inter-cluster dispersion and intra-cluster dispersion. Higher values indicate better-defined and more separated clusters [26].

All this evaluate metric were useful for comparing the behavior and effectiveness of K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models on the housing dataset. Also to numerical evaluation, PCA visualizations were also used to better understand cluster distributions and relationships between observations.

B. Clustering Results

We applied multi clustering algorithms to the preprocessed housing dataset after feature scaling and outlier treatment. Principal Component Analysis (PCA) was used to visualize the generated clusters in two-dimensional space.

As we find in the result K-Means produced relatively balanced and compact clusters with clearer separation boundaries compared to other algorithms. As shown in Fig. 1, the clusters generated by K-Means appear well distributed across the PCA space, indicating effective partitioning of the dataset.

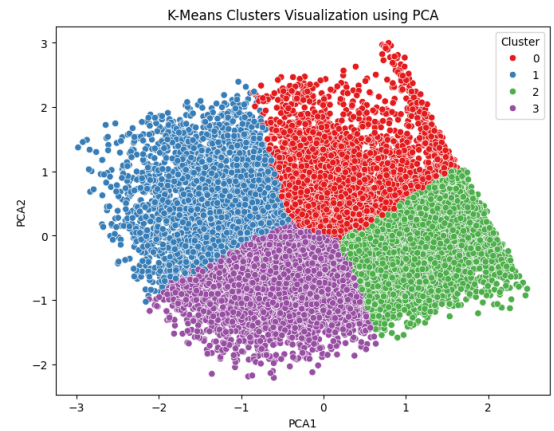


Fig. 1: K-Means clustering visualization using PCA.

In the result final we find that DBSCAN generated one dominant dense cluster along with several small clusters and noise points. Approximately 1263 observations were classified as noise, representing around 7.5% of the dataset. Fig. 2 illustrates the ability of DBSCAN to identify sparse observations and outliers. However, the visualization also shows that DBSCAN struggled to produce strongly separated density-based clusters for this dataset.

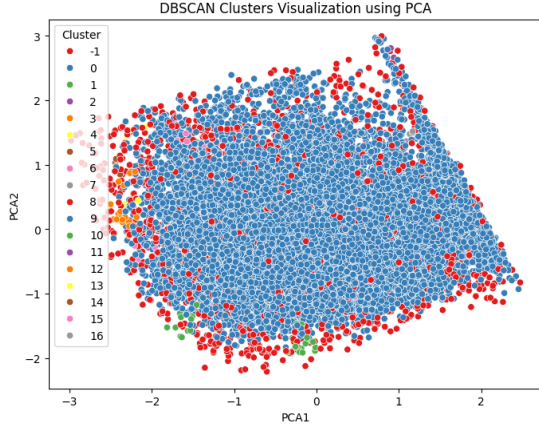


Fig. 2: DBSCAN clustering visualization using PCA.

In other hand, the Agglomerative Clustering successfully grouped the observations into hierarchical clusters. The PCA visualization in Fig. 3 reveals meaningful separation between groups. In addition, the dendrogram shown in Fig. 4 illustrates the hierarchical merging process and relationships between clusters.

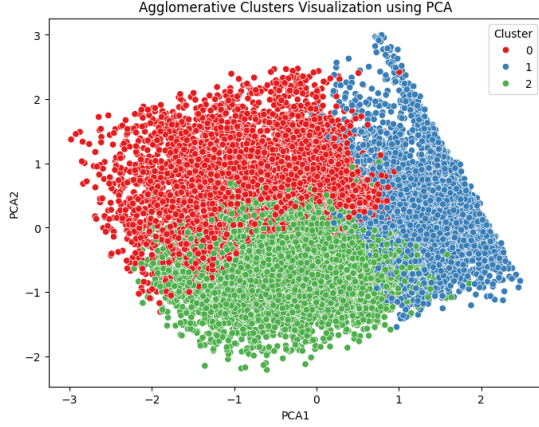


Fig. 3: Agglomerative clustering visualization using PCA.

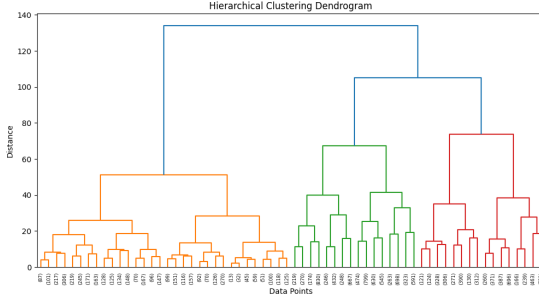


Fig. 4: Hierarchical clustering dendrogram.

The last clustering algorithm which is Gaussian Mixture Models (GMM) produced flexible cluster boundaries and handled overlapping observations more effectively. As shown in

Fig. 5, GMM generated smoother transitions between clusters because of its probabilistic clustering mechanism.

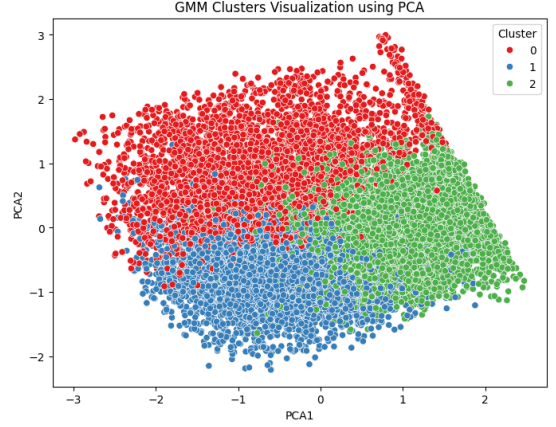


Fig. 5: Gaussian Mixture Model clustering visualization using PCA.

C. Comparative Analysis

Table I summarizes the comparative analysis of the implemented clustering algorithms.

TABLE I: Comparative Analysis of Clustering Algorithms

Algorithm	Cluster Separation	Outlier Detection	Complexity
K-Means	High	No	Low
DBSCAN	Medium	Yes	Medium
Agglomerative	High	Limited	High
GMM	High	Limited	Medium

The results of this study showed that K-Means and GMM achieved the most visually balanced clustering structures on the housing dataset, while DBSCAN was effective for noise detection but struggled to create strongly separated density-based clusters because of the continuous nature of the housing data. In other hand agglomerative Clustering provided meaningful hierarchical relationships but required higher computational resources.

D. Critical Discussion

We concluded from the comparative analysis that clustering performance strongly depends on data characteristics and pre-processing techniques. Feature scaling significantly improved clustering quality for distance-based algorithms such as K-Means and GMM.

The study also highlighted the importance of parameter tuning. In DBSCAN, inappropriate values of the *eps* parameter resulted in either excessive noise detection or the formation of a single dominant cluster. This confirms the sensitivity of density-based clustering methods to parameter selection.

Using Interquartile Range (IQR) method for Outlier treatment shows big impact for clustering stability and reduced the impact of extreme observations, while PCA visualization also proved useful for understanding cluster distributions and comparing the behavior of different algorithms.

In the big picture, the K-Means algorithm and Gaussian mixture models delivered the best performance in data aggregation for the housing dataset, while the DBSCAN algorithm demonstrated superior noise detection capabilities. Agglomerative Clustering offered valuable hierarchical insights but required higher computational cost.

We can explain the behavior of DBSCAN in this study by the inappropriate parameter values, which caused the algorithm to form a single dominant cluster, showing its high sensitivity to parameter tuning. However, the algorithm successfully detected noise points and sparse observations. This study also agrees with previous research findings [6].

E. Comparison with Previous Studies

What we found in this study mostly lines up with what other clustering research has shown before. For instance, K-Means created neat, fairly distinct groups once we cleaned up the data and scaled the features, much like that study presented in [5]. This method also proved to be stable and pretty fast when we used it on the housing data.

Agglomerative Clustering's results were pretty similar to what other hierarchical clustering studies have found too [6]. This method showed us how the data points related to each other in a meaningful, layered way, and the groups it formed looked a lot like what other comparison studies have reported.

Also, how well Gaussian Mixture Models (GMM) performed backed up what the JORSClustering study concluded [7]. Compared to methods like K-Means, which just uses centers for groups, GMM was more flexible. It used probabilities to form clusters and gave us smoother lines between them. When we looked at the data using PCA, it was also clear GMM could handle situations where data points from different groups slightly overlapped.

So, all in all, our findings here confirm what earlier research has suggested: how well clustering works really depends on how well you prepare your data, what kind of data you have, and how you chose the settings.

F. Limitations of the Study

Although this study achieved satisfactory results, several limitations should be considered. First, clustering performance strongly depended on preprocessing quality and parameter selection. Algorithms such as DBSCAN were highly sensitive to the choice of the *eps* parameter and the minimum number of samples, which directly affected cluster formation and noise detection.

Another limitation is related to the dataset itself. The housing dataset we used in this work contained only numerical socioeconomic and demographic variables. Additional features or larger datasets may potentially improve clustering diversity and provide more representative grouping structures.

In addition, we used PCA only for two-dimensional visualization purposes. Although PCA simplified the interpretation of cluster distributions, dimensionality reduction may lead to partial information loss and may not perfectly preserve all relationships between observations.

On the other hand, Computational complexity also represented a limitation for some algorithms. Agglomerative Clustering required more computational resources compared to K-Means and DBSCAN, especially when processing larger datasets.

Finally, this study mainly relied on internal clustering evaluation metrics and visual analysis. As future work, this study can be extended by testing additional clustering algorithms such as Spectral Clustering, OPTICS, or Mean Shift. Future research may also include the use of larger and more complex datasets, advanced dimensionality reduction techniques, and automated hyperparameter optimization methods to further improve clustering performance and visualization quality.

VI. CONCLUSION

In this study, we presented a comparative analysis of several unsupervised clustering algorithms applied to a housing dataset containing socioeconomic and demographic characteristics. Our main objective in this work was to evaluate and compare the behavior and performance of different clustering techniques on the same dataset using preprocessing methods, internal evaluation metrics, and PCA visualization.

Before applying the clustering algorithms, we performed many preprocessing steps which included data cleaning, feature scaling using the *StandardScaler* method, and outlier treatment using the Interquartile Range (IQR) method. These preprocessing techniques have contributed significantly to improving clustering quality and model stability.

We used four clustering algorithms which were implemented and compared: K-Means, DBSCAN, Agglomerative Clustering, and Gaussian Mixture Models (GMM). This study results showed that K-Means and GMM generated more balanced and visually separated clusters, while Agglomerative Clustering provided meaningful hierarchical relationships between observations. On the other hand, DBSCAN demonstrated strong capabilities in detecting noise points and sparse observations but showed high sensitivity to hyperparameter tuning.

After we compare between these clustering algorithms we found that clustering performance strongly depends on dataset characteristics, preprocessing quality, and algorithm parameter selection. In addition, PCA visualizations proved useful for understanding cluster distributions and comparing the behavior of different clustering methods.

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